Giving AI Assistants a Reliable Notebook

How Agent Memory helps AI keep track of long, multi-day work

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The short version

AI assistants are increasingly asked to do work that takes many sessions, such as writing software or working through a long derivation. But they usually start each session with a blank mind. Existing fixes either pile up so much text that the important parts get lost, or they *summarize* and lose exact details.

We built **Agent Memory**, which works like a well-kept lab notebook: it saves whole notes, tracks their history, hands the assistant a one-paragraph cheat sheet each morning, and lets it look up the full page when needed. On a hard physics benchmark, it kept every needed equation available and helped a strong AI model finish **all 20 multi-step problems**, compared with 7 without it.

1 The problem: a colleague who forgets every morning

Imagine a brilliant colleague who gets amnesia overnight. On Monday you explain the setup of a project. On Wednesday you ask them to build on Monday's result. They remember nothing.

That is roughly how AI assistants work today. Each new session starts fresh. To work around this, people give the assistant some kind of memory. The common approaches all have a weakness:

- **Hand over the whole transcript.** Nothing is lost, but the useful facts are buried under dead ends and chatter, and the cost grows with every session.
- **Keep short summaries.** A tool writes brief notes such as "the system has a rule about signs." The exact rule, the number, the sign, is the part that gets lost.
- **Search the transcript in fixed-size slices.** The text is cut every few hundred characters. A definition and the result that depends on it can end up in different slices, so the assistant finds one without the other (Figure 1).

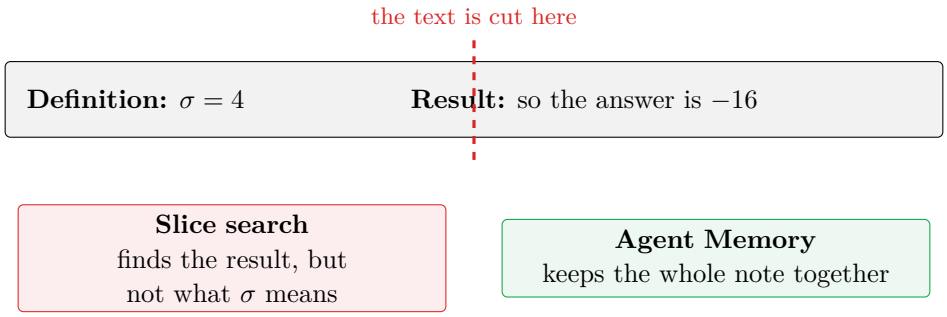


Figure 1: Cutting text into fixed slices can separate a definition from the result that depends on it.

2 The idea: a good lab notebook

Scientists and engineers solved this long ago with a notebook. You write full entries. You can see when something changed. And each morning you glance at a one-page summary before diving back in. Agent Memory gives AI assistants those same habits.

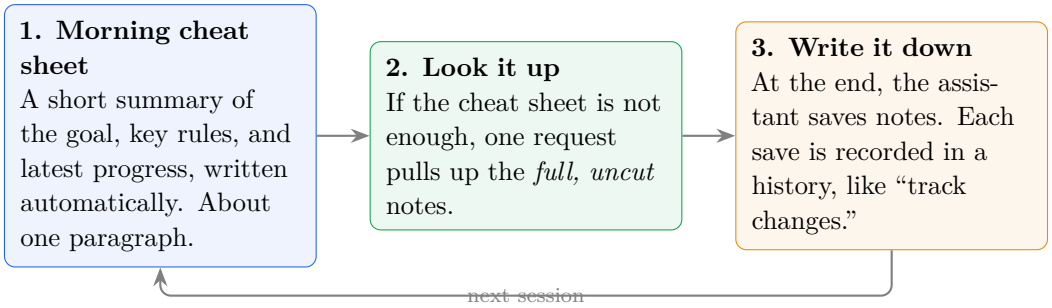


Figure 2: The three steps of Agent Memory, repeated every session.

Four design choices, in plain words

Keep whole notes. Agent Memory never rewrites an equation into a vague summary. Longer notes can be folded into a short roll-up, but the originals always stay available.

Remember the history. Every note is saved with a snapshot of the project's files, like a version history. If a file changes after a note was written, the note is flagged as *possibly out of date*. The assistant can even step back to how things looked when a note was written.

Use a fixed-size cheat sheet. The morning summary is capped at about 1,200 characters, roughly a short paragraph. It is built by simple rules, with no extra AI call, so it is fast, cheap, and predictable. It always leads with the goal.

Search that understands symbols. Ordinary search engines stumble on things like $[D, K]$ because they treat punctuation as part of the word. We fixed that, which raised the chance of finding the right equation in one test from 29% to 100%.

3 How we tested it

We used *MemoryArena*, a public test designed for exactly this problem. It contains 20 theoretical-physics problems, each broken into a chain of 2 to 12 steps. Step 1 explains the setup. Every later step starts with a blank mind and depends on earlier answers. That makes it a good stress test for memory: the assistant cannot succeed by being clever alone, it also has to remember.

We measured two things:

- **Was the information still available?** For each step, how many of the needed equations were included in what the assistant was shown?
- **Did the assistant finish the problem?** How many of the 20 chains ended with a correct final answer?

4 What we found

Finding 1: the exact math survives

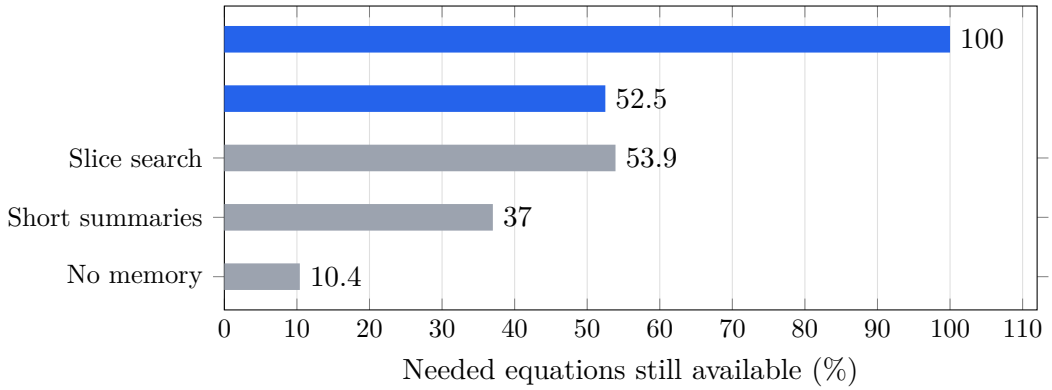


Figure 3: Short summaries lose most of the exact math. Slice search loses almost half. The cheat sheet alone is on par with slice search, and the full lookup keeps everything.

Finding 2: a mid-sized AI finishes more problems

We first used the benchmark's standard model (`gpt-5-mini`). On its own it gets only 12 of the 20 first steps right, so there is a ceiling on what any memory could achieve.

Of the 12 problems this model could solve at step 1, it finished 10 (83%) with Agent Memory.

Finding 3: a stronger AI finishes all of them

To separate memory quality from raw ability, we repeated the test with a stronger model (`gpt-5.6-sol`) that gets all 20 first steps right. Now any mistake is a memory mistake.

The cheat sheet did not suffice on the longest chains, since a paragraph cannot hold twelve steps of algebra. That is exactly where the one-request full lookup helped.

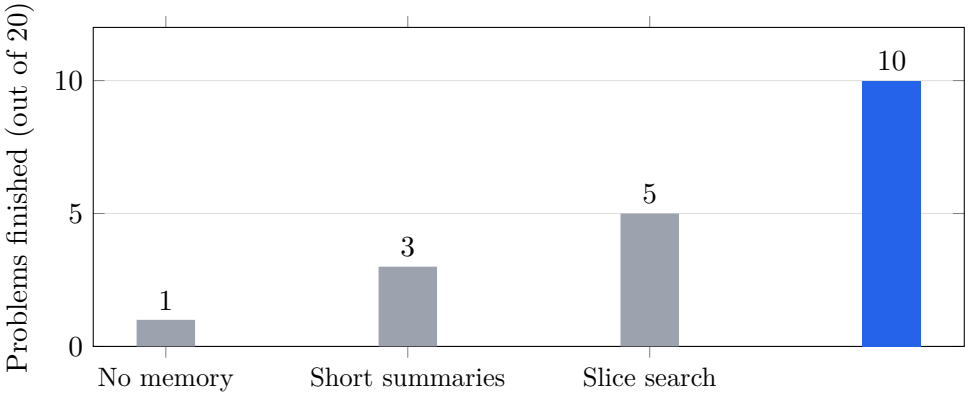


Figure 4: With the standard benchmark model, Agent Memory finished 10 of 20 problems, about five times as many as no memory and twice as many as slice search.

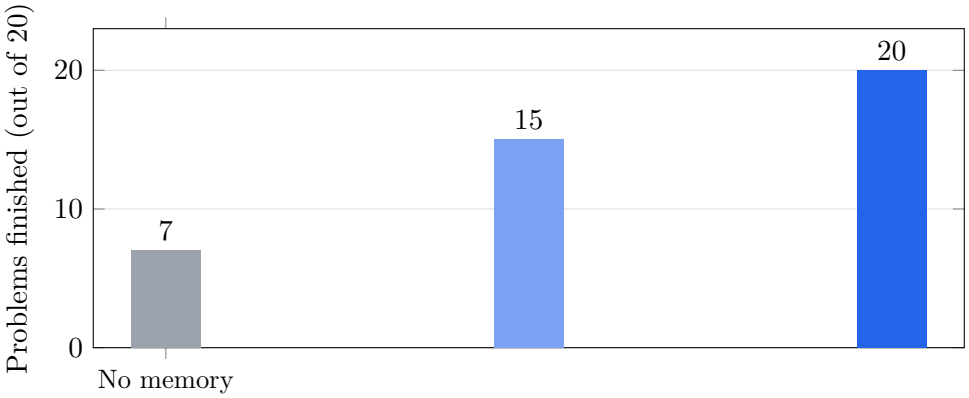


Figure 5: With a strong model, the one-paragraph cheat sheet alone finished 15 of 20 problems, and adding one full lookup finished all 20 (86 of 86 steps).

Finding 4: the effect grows with the length of the task

Without memory, even the strong model's success rate fell from 100% on step 1 to 45% on step 2, and kept falling. With the full lookup it stayed at 100% even at step 12.

5 Beyond physics: remembering rules

Agent Memory can also turn a pile of repeated examples into a short rule. In one test, three similar review notes were merged into a single rule and then generalized. The stored rules shrank by about 65% and the summary shown to the assistant by 50%, while the complete history stayed in the record. The rule engine is deliberately restricted so it can only do simple, safe checks and cannot run arbitrary code.

6 What this does and does not show

We want to be clear about the limits of these results.

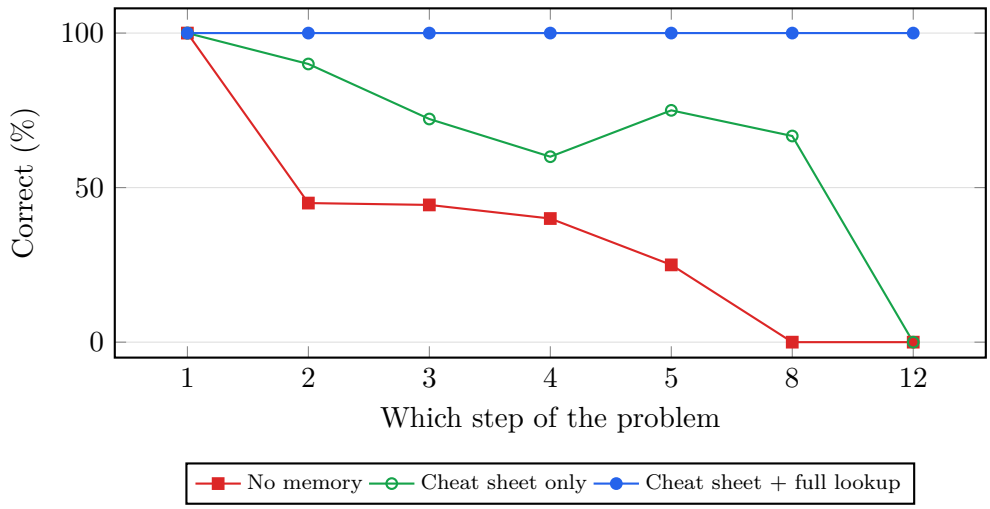


Figure 6: Accuracy by step with the stronger model. Without memory it decays quickly; with the full lookup it does not.

- **One benchmark, one field.** All results are on physics derivations. We have not yet tested the other MemoryArena areas, such as shopping or travel planning.
- **Small test.** Twenty problems means each one is worth 5 percentage points, so small differences should not be over-read.
- **Tuned on the test.** We found and fixed the search problems by studying failures on this same benchmark. Fresh problems are needed to confirm the results hold up.
- **Not the top score overall.** Some published memory systems that use other models scored as high or higher on this benchmark. Differences in setup mean these numbers are not a fair head-to-head, so our claim is about the exact-math problem above, not about being the best overall.
- **Long chains need the lookup.** One paragraph is not enough for very long problems.

7 Conclusion

Memory for AI does not need to be exotic. Keep complete notes, record their history, show a short fixed-size summary each time, and allow one lookup of the full page. On a demanding test where exact details matter, this kept the math intact and let a strong assistant finish every problem. The code is open at the link above, and the full technical write-up has the details.

Glossary. **Agent:** an AI assistant that works on a task over many steps. **Session:** one work period; the assistant starts each with a blank mind. **Long context:** giving the AI the entire past transcript. **RAG (retrieval-augmented generation):** searching stored text and showing the AI the best matches. **Git:** a version-history tool used by programmers, like "track changes" for a whole project. **Benchmark:** a standard test used to compare systems.